Skin Cancer Dataset: SQL Analysis

SQL Capstone | Hyginus Echezona Okonkwo | 31 October 2026

1. Objective and workflow

Analyse the supplied patient and lesion data using PostgreSQL to describe diagnosis patterns, demographic and environmental associations, and lesion characteristics, and prepare joined metadata for subsequent research.

The supplied PostgreSQL backup was restored into a dedicated database. The SQL script was executed in pgAdmin against PostgreSQL 18. Patient data (public.table1) and lesion data (public.table2) were joined on patient_id through a temporary analytical view. Source tables were retained unchanged. Query execution is evidenced by the supplied pgAdmin screenshots; CSV-derived figures were independently checked from the exported file.

2. Data validation and preparation

Check	Result
Patient rows / lesion rows / joined rows	1,088 / 1,088 / 1,088
Unmatched lesion rows	0
Unique patient IDs in CSV	1,088
Repeated lesion ID values	81; each occurs under two patients
Exact duplicate rows in exported CSV	0
Missing skin type after zero conversion	579 (53.2%)
Missing diameter pairs after zero conversion	581 (53.4%)

The brief states 1,089 instances; the supplied restored data and export contain 1,088. The reason for this difference is unconfirmed. All records were retained. Repeated lesion IDs are not sufficient evidence of duplicate observations: the affected records belong to different patients. The patient_id/lesion_id pair is unique in this export and can track records pending clarification of the original ID convention.

Age, gender and diagnosis contain no nulls. The executed quality query found no negative diameters. Zero skin types and diameters were treated as unavailable rather than measured values; their original meaning should be confirmed with the data provider.

3. Diagnosis and age patterns

Diagnosis code	Records	% of all records
ACK	461	42.4%
BCC	273	25.1%
NEV	144	13.2%
SEK	137	12.6%
SCC	56	5.1%
MEL	17	1.6%

For descriptive comparisons, malignant_proxy is true for BCC, SCC or MEL: 346 records (31.8%). All other diagnosis codes form the comparison group; they should not all be described as benign. These are dataset labels, not new diagnoses made by this analysis.

Age group	Records	BCC/SCC/MEL	Within-group %
Under 30	57	2	3.5%
30-49	221	49	22.2%
50-69	513	174	33.9%
70+	297	121	40.7%

The proportion labelled BCC/SCC/MEL increases across age bands. These are proportions among supplied records, not population incidence or individual predicted risk.

4. Demographic and environmental comparisons

Factor	Group	Records	BCC/SCC/MEL %
Gender	FEMALE	362	43.4%
Gender	MALE	726	26.0%
Smoking	False	1026	29.1%
Smoking	True	62	75.8%
Alcohol	False	950	26.1%
Alcohol	True	138	71.0%
Pesticides	False	865	22.0%
Pesticides	True	223	70.0%
Prior skin cancer	False	864	21.9%
Prior skin cancer	True	224	70.1%

Factor	Group	Records	BCC/SCC/MEL %
Family cancer history	False	816	19.5%
Family cancer history	True	272	68.8%

5. Symptoms and body region

Feature	Absent: n / labelled %	Present: n / labelled %
Bleeding	866 / 19.2%	222 / 81.1%
Change	989 / 28.6%	99 / 63.6%
Elevation	477 / 12.6%	611 / 46.8%
Growth	578 / 15.1%	510 / 50.8%
Itching	418 / 20.6%	670 / 38.8%
Pain	937 / 23.3%	151 / 84.8%

Pain and bleeding show large unadjusted differences in diagnosis proportions. These comparisons do not measure diagnostic accuracy, sensitivity or specificity and should not be used as a standalone clinical decision rule.

Body region	Records	BCC/SCC/MEL	Within-region %
FACE	278	102	36.7%
FOREARM	219	25	11.4%
CHEST	124	52	41.9%
BACK	105	38	36.2%
ARM	92	23	25.0%
NOSE	68	47	69.1%
HAND	55	7	12.7%
NECK	37	20	54.1%
THIGH	35	9	25.7%
EAR	32	17	53.1%
ABDOMEN	20	2	10.0%
SCALP	11	0	0.0%
LIP	7	4	57.1%
FOOT	5	0	0.0%

The face has the most records. Small regional groups, such as lip and foot, give unstable percentages; zero observed cases does not establish zero risk.

Infrastructure variables

The executed SQL also compared piped water and sewage access. The visible results show BCC/SCC/MEL proportions of 63.7% with piped water versus 19.3% without. This is an unadjusted sample association.

Infrastructure fields are absent from the Section 10 CSV and were not independently recalculated from that export. Such differences require checking coding, sampling and confounding before interpretation.

6. Lesion size

Diagnosis	Valid pairs	Mean size (mm)	Median size (mm)
ACK	133	9.1	8.0
BCC	271	10.0	9.0
NEV	16	6.1	5.5
SEK	14	7.8	6.5
SCC	56	10.5	8.8
MEL	17	14.1	13.0

Size is the arithmetic mean of the two recorded diameters, calculated only when both are available. MEL has the largest mean in this sample. Missingness differs markedly by diagnosis: valid pairs range from 14/137 for SEK to 17/17 for MEL. Size comparisons may therefore be biased.

7. Biopsy coverage and exported metadata

Executed SQL results show 458 records marked biopsied (42.1%): 273 BCC, 56 SCC, 17 MEL, 95 ACK, 12 NEV and 5 SEK. All 346 BCC/SCC/MEL records are marked biopsied. Biopsy status is metadata; the underlying pathology reports and label provenance were not reviewed.

The exported CSV contains 1,088 rows and 21 columns. It retains patient and lesion IDs for audit and traceability, excludes biopsy status and image filenames, and converts zero-coded measurements to nulls. The script comment claiming identifiers are excluded is inconsistent with its SELECT statement: IDs are retained in the export and must be excluded from model predictors. diagnostic and malignant_proxy are alternative target labels; neither should predict the other. The export is prepared metadata, not a trained or validated model.

8. Recommendations

1. Confirm the record-count discrepancy and lesion ID convention with the data provider; retain original records and use the unique patient/lesion pair for traceability.
2. Improve recording of skin type and both diameters, and investigate why measurement completeness varies by diagnosis.
3. Investigate the observed age, symptom and exposure associations with adjusted analysis and suitable independent data before drawing clinical or causal conclusions.
4. For future modelling, verify labels, address missing values within training folds, encode categorical variables, exclude IDs and biopsy status from predictors, and assess class imbalance. No model training is required by the explicit SQL workflow.

9. Limitations and conclusion

The analysis is descriptive and unadjusted. Sampling, recruitment and label provenance are not established; findings cannot be generalised to the population or interpreted causally. Repeated lesion IDs and substantial, diagnosis-dependent missingness limit interpretation. Some small groups contain few observations. SQL joins, summaries and quality checks address the brief, and the export supports further preparation. The most frequent label is ACK; BCC/SCC/MEL proportions vary by age, exposure and lesion features. Reliable further research depends on clarifying identifiers and improving measurement completeness.